



Framework for Automatic Semantic Annotation of Images Based on Image's Low-Level Features and Surrounding Text

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Abstract

Semantic annotation of images is the process of assigning metadata in the form of captions to a digital image. This is an important process for the indexing and searching of images in a big database. In this paper, we present the framework of automatic semantic annotation of images and explore the effectiveness of using it to annotate images based on both the image's low-level features and its surrounding text. In the proposed framework, the image's features have been extracted by using convolutional neural networks, while words in the surrounding text have been represented by word-embedding vectors. Both modalities are further processed using recurrent neural networks with long short-term memory cells that possess an attention mechanism to generate an annotation sentence that describes the image. Empirical evaluations of the proposed framework, acquired using a news dataset, show promising performance results and are comparable to the results of recent image annotation systems. The produced semantic annotations in free-text format can be further converted into a structured resource description framework that enables more expressive queries across a diverse source of images.

Keywords Image processing · Feature extraction · Semantic annotation

1 Introduction

People generate many documents with pictures and store them on the web every day. In such big data, indexing and searching techniques become a critical need to retrieve desired images quickly. Until now, systems that are retrieving desired images based on semantic annotation are not satisfying the needs of the users. One approach to retrieve the desired images is to annotate images with tags or to give metadata to all images so that we can use a text-based query to retrieve the images later on. However, annotating a huge archive of images automatically is yet a goal as doing this task manually is almost impossible. Most automatic image annotation systems rely on machine learning techniques to let the computer learn patterns from training images because there is no such exact mapping between patterns of brightness values in an image matrix and semantic concepts. However, putting only raw images as input to a machine-learning model is not

enough, as raw images contain too much redundancy and noise that likely hide the information, and hence the model may not learn precise patterns [1, 2]. Because of this problem, we need feature-extraction techniques to compress the data more compactly than the raw data, such that we remove redundancy while retaining relevant information. This will ease the classification task because the data pattern can be more easily discovered.

One of the most popular feature-extraction methods for image annotation is the bag-of-visual-words as image representation [3], which is a histogram of local key points in the image. This approach involves a method of several hand-crafted features, including corner detection algorithm and scale invariant feature transform (SIFT) [4–7]. On the other hand, deep learning-based features, which are one type of learned features or representation learning methods, are gaining much popularity [8–11]. Deep learning is a relatively new area, but it has been gaining a lot of interest in the communities of researchers on machine learning and computer vision. One reason is that to a certain extent, the feature-extraction techniques are independent of any particular classification task, so they can be used for various applications. In addition, deep-learning-generated features can generally outperform

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the existing popular features-extraction techniques. In addition to the image's features, one can make use of the surrounding text of images to generate semantic image annotation. Surrounding text can help in adding information that is not available or difficult to extract from the image. One way to extract textual information is to use word-embedding features and sequence-to-sequence recurrent neural networks (RNN), which are used in text summarization and natural language translation.

In this paper, we present the framework of automatic semantic annotation of images (ASAI), leveraging deep learning methods in computer vision and natural language processing (NLP) to generate semantic image annotation, which helps to retrieve the desired images quickly. Moreover, image annotations may also help visually impaired people to understand the image through text to voice conversion system [12]. Deep learning techniques help the framework to create annotations without utilizing many human-generated resources, such as rules or hard-coded templates, and rely on training data instead. The contribution of this paper is to generate semantic image annotations by taking advantage of the information contained in the image's pixels and the surrounding text using new techniques, namely Convolutional Neural Networks (CNN) and RNN. A natural language description would be able to express the relations among objects depicted in the image instead of a list of words. The surrounding text is used to extract further information that may not be available from the image pixels. Furthermore, we present a dataset that is suitable for generating image annotations based on image pixels and text, as we do not find a suitable dataset to assess the proposed framework. The dataset is acquired by filtering data from the existing dataset ION [13] to suit the dataset to our task. Our dataset contains approximately 80,000 news articles with images and captions. The rest of this paper is structured as follows. Section 2 reviews the related work. Section 3 describes the proposed semantic annotation model. Section 4 presents the experimental results and discussion. Finally, Sect. 5 concludes the paper and discusses future work.

2 Related Work

In this section, we review the work related to the area of semantic image annotations. First, we summarize several works that create a set of words to describe objects in a given image. Then, we review other works that create alternative forms of annotations instead of a set of words. We also discuss previous methods for extracting information from text, as we want to use both image and textual features to generate semantic image annotations. Finally, we summarize existing image annotation systems, along with some challenges and issues regarding them, which motivated us to do this research.

2.1 Image Object Recognition

Over many years, object recognition has relied on hand-crafted features such as SIFT [4]. Perhaps one of the most successful feature-representation of images is based on the bag of words pipeline, which has been dominantly used in recent works [14–17] and in several popular challenges such as PASCAL VOC [18]. A bag of visual words is a collection of unordered occurrences of interesting patches in an image [3] encoded by local feature extraction such as SIFT. As an illustration, an image depicting a motorbike may have a high occurrence of image patches of handlebars and wheels, whereas non-motorbike images may not. We classify images by feeding this histogram of interesting image patches to some machine learning classifiers such as SVM.

Recently, we see a lot of work using CNN to do image object recognition. CNN is one of the deep learning-based methods, and it is gaining a lot of popularity with the computer vision researchers' community. CNN can give good results and outperform many object classification approaches that use hand-engineered feature-extraction techniques. As reported in the ImageNet yearly competition [19], the top performers have been using CNN since 2012, when Krizhevsky et al. [20] used CNN for the first time in the competition. Based on their work recently, CNN can be improved by increasing the number of hidden layers [9] and modified for object detection [21].

The disadvantages of the deep learning-based approach are that it needs much parameter tuning, a large training dataset, and a long training time. The large dataset problem could be addressed by making use of data that can be easily obtained from the Internet and crowdsourcing. The long training time could be reduced by using GPU parallel processing [10].

2.2 Generating Free-Text Descriptions of Images

Sentences are richer than a list of words because they describe properties of objects, relations between entities, and activities. Therefore, some works try to generate free text from images instead of a list of tags or keywords. An image description system can help people better manage the increasing volumes of multimedia data. Such a system would save much human labor and provide people with easier access to large-scale multimedia resources, i.e., automatic image caption generation could assist journalists in creating descriptions for the news pictures or videos associated with their articles.

Recent famous works to generate image descriptions use a variety of RNNs. An RNN is a type of Neural Network (NN) where connections between neurons form a loop. This creates an internal state of the NN that retains a memory of the previous sequence input. This internal memory can



be used to process arbitrary sequences of inputs and output. Therefore, an RNN can handle multiple forms of input and output, rather than a fixed-sized input. This makes the RNN applicable to tasks that need a sequence of outputs such as generating image descriptions. However, there is a problem with the RNN called vanishing gradient. RNN loses information in an earlier sequence when the number of input sequences gets long. The gradient used in training becomes so small it makes the training too slow. To address this problem, Hochreiter et al. introduced Long short-term memory (LSTM) network [22]. LSTM uses three gates instead of a single neuron at each time step to control the information flow between input sequences. These gates prevent the gradient-vanishing problem during training and make LSTM generally perform better than a basic RNN.

One recent work that uses RNN to generate sentences from an image is by Karpathy et al. [23]. The most powerful representation of images and text carried out by deep learning-based approaches is namely CNN for generating image features and word embedding for creating text features. Based on these features, Karpathy et al. [23] tried to produce a free-text description of an image. In their work, image regions' features are extracted using RCNN proposed by [21], while text description is mapped based on word embedding and the Stanford dependency parser. Multiple instance-learning algorithms are used to minimize a cost function, defined as the degree of matching between image regions and their text description alignment. Later on, they refined their work by introducing bidirectional RNN for extracting and aligning text description features, instead of using Stanford parser and Multi-instance learning [11].

Better results were achieved by Vinyals et al. [24] using LSTM networks. In the proposed framework, to generate a sentence description of an image, the LSTM first takes the image features produced by CNN [25] as a first sequence. Then, the LSTM generates word probabilities that occur at the beginning of the sentence. We take the word that has the highest probability into the word-embedding-feature vector [26], and feed it to the LSTM again as the next input sequence. This is repetitively done until we get a unique word that indicates the end of the sentence.

2.3 Using Surrounding Text for Image Annotation

Some works have used surrounding textual and image features to annotate images [1, 2]. Most of the existing approaches for image annotation generally demand hand-labeled training data, which are limited and expensive to obtain. Associated text or surrounding text, which is more abundant, could improve the annotation system. Surrounding text may contain information that cannot be mined from images. However, the text could also be noisy, i.e., irrelevant

to the image, so a sort of noise-reduction technique should be used.

Ding et al. [27] used words in surrounding text to label images. First, they build training data by gathering images from image search engines such as Google and Yahoo using a keyword-based query from LSCOM ontology. Every image in the search results was labeled using words in its surrounding text that were similar to those used in the keyword-based query. After building the training data, the system annotates a target image by comparing it to images in the training dataset, then annotations from images that are visually most related will be used to label the target image.

Latent Dirichlet Allocation (LDA) [28] is a method to leverage both visual and textual modalities. LDA is used to assign suitable topics for a given document. Feng et al. [29] proposed a probabilistic relevance model that captures the joint probability of annotated words and images. After creating several keywords, LDA is used to filter out irrelevant annotations. The authors claimed that their model handles a noisy dataset using the joint probability of word and image features. Feng [30] proposed an approach that differs from [29]; it relies on LDA to infer relevant topics that capture the co-occurrence of visual features and words. Feng et al. [31] introduced a model that can generate captions or sentence descriptions given an image and its text description. They used LDA to generate topic keywords as annotations, and the N -gram model to generate sentences from a list of keywords. However, the generated sentences could still have grammatical and semantic errors.

Tian et al. [32] extended previous LDA works by introducing “salient keywords” as additional features to be combined. Salient keywords refer to words in the surrounding text that directly describe the salient objects in the images.

2.4 Text Summarization

Text summarization is the task of generating a short text, based on the input while maintaining the main information from the original. This task is akin to the image annotation task using surrounding text because both of them make use of the information in the text input. Without access to the text input, our task in this paper would be identical to the natural image caption generation task addressed by Karpathy et al. [11, 33], and Vinyals et al. [24]. Earlier work on text summarization focused on extractive summarization, which is generating a summary by taking a subset of a sentence from the input. Helmy [34] proposed an image annotation framework with a hierarchical classification approach that consists of four annotation levels. The proposed framework achieved a significant improvement in the categorization rate that leads to a more accurate annotation of images.

Daumé et al. [35] and Vanderwende et al. [36] suggested that a document's content can be viewed as a probabilistic



combination of pre-defined topics. Every word in a document is associated with a topic with some probability. This topic distribution is modeled by LDA [28], which is similar to probabilistic latent semantic analysis (PLSA) or probabilistic latent semantic indexing (PLSI) [37]. The sentences with topic distributions highly similar to the whole document are selected as the summary. As gathering data has become easier and inspired by the recent success of deep neural networks on machine translation, recent works have utilized the data-driven approach for generating abstractive text summarization. Unlike extractive summarization, abstractive summarization generates output sentences that do not have to be a part of the input, while maintaining the main information. This is considered more challenging than extractive summarization because it tries to make brand-new sentences that do not appear in the input. The generated sentence may completely change to make concise and informative summaries.

One of the popular models on abstractive text summarization is Rush et al. [38]. Their work is based on Bahdanau et al. [39] who proposed neural machine translation with encoder-decoder family and Sutskever et al. [40]. The NN in their approach consists of two parts (encoder and a decoder). The encoder reads an input sentence and encodes it into a fixed-length vector. The vector reads by the decoder to generate a translation sentence output of the source sentence. Both parts are jointly trained to maximize the probability of a correct translation given a source sentence. A potential problem with this approach is that the encoder part compresses all the input sequences. This may make it difficult for the encoder to keep the main information, especially when the input has very long sentences.

Cho et al. [41] showed the decreasing performance of a basic encoder-decoder as the length of the input increases. Bahdanau et al. [39] proposed the attention mechanism that allows the decoder to attend every state in the encoder before generating each step of the output sequence. Each time the model generates a word in an output sequence, it looks for a set of words in the source sentence that contain relevant information by computing the context vector. The context vector contains a set of weights between the output and all the input showing the relevancy or importance of the input. The model then predicts a target word based on the last state vector of the decoder and the context vector.

We used the attention mechanism as a part of the proposed framework. However, the attention mechanism uses too much memory when dealing with long input sequences, because the attention vector computation needs every state of the encoder for each input in the sequence to be stored in memory. Therefore, we have to limit the input articles by taking only small numbers of first sentences in the input in our experiments.

Based on the intensive survey, we highlighted some challenges and issues related to the image annotation systems as follows. Some solutions are problem specific and cannot apply to other domains. We found limited works that use noisy data. We found limited work that leverages both textual and content-based image features. RNNs are gaining popularity, but their applications for combining textual and content-based features to generate image annotations are limited. Some early work in image annotation focuses on how users can gather further or related information based on a query using ontologies and manual annotation, while recent works try to automate annotation processes using various machine-learning techniques. CNN is gaining much popularity as an end-to-end solution, from image-feature extraction to image labeling, while the RNN is popular in NLP tasks. However, these still need to be extended to address more real-world problems. Recent works in image caption generation use a simplified dataset where every sentence caption relates to its corresponding image. Obtaining a good dataset may not always be easy, so gathering easily crawled data on the Internet should be considered. In this paper, we explore the effectiveness of using content-based and text-based features from surrounding text to generate semantic image annotations using RNN with a dataset that is obtained without much time and human resources. Relying on content-based features only to produce semantic image annotations may not be enough, as the content-based approach cannot capture abstract concepts. On the other hand, surrounding context-based features can provide abstract concepts but can be noisy as well and may give irrelevant information to the images. We show that noisy data can be addressed by leveraging both content-based and text-based features.

3 Automatic Semantic Annotation of Images Framework

In this paper, we present the ASAI framework and explore its feasibility of generating semantic image annotations from images' pixels and the surrounding text. As a part of the proposed framework, we propose a long short-term memory (LSTM) neural network-based model to carry out the semantic annotation task. The LSTM model is intended to maximize the probability of correct sentence description, given the image and surrounding text based on Eqs. 1 and 2:

$$\theta^* = \arg \max_{\theta} \sum_{(I, A, S)} \log p(S|A, I; \theta) \quad (1)$$

$$S^* = \arg \max_S \sum_{(I, A, S)} \log p(S|A, I; \theta^*) \quad (2)$$

θ^* represents the optimal parameter of the model, S^* is the predicted sentence description, S is the correct sentence description (based on the ground truth), I is an image, and A is the surrounding text of the image. Both S and A are a sequence of words $S_0, S_1, \dots, S_N$ and $A_0, A_1, \dots, A_M$ respectively, where N is the last index sequence of the predicted sentence and M is the last index sequence of the input article. Using the probability chain rule, the probability of a predicted sentence given the input article and image is computed from Eq. 3:

$$\log p(S|A, I) = \sum_{t=0}^N \log p(S_t|I, A_0, A_1, \dots, A_M, S_0, S_1, \dots, S_{t-1}) \quad (3)$$

As shown in Fig. 1, the ASAI framework comprises LSTM RNN. This LSTM neural network needs inputs from images under processing with CNN, and their surrounding text, handled by a word-embedding model. In general, the ASAI framework does the following steps that are explained in the next sub-sections:

- It accepts the input image and its surrounding text.
- It extracts image features using CNN.
- It preprocesses the surrounding text.
- It gets a sequence of word tokens from the text-preprocessing step.
- It translates each word token to its word feature vector.
- The extracted text and image features feed sequentially into the LSTM neural network. The first sequence is the feature of the first word appearing in the surrounding text. The last input sequence will be the special token “end of the sequence.”
- During training, the final output is compared to the image caption in the news website to compute the cost function.
- The weights are then updated by minimizing the cost function.

3.1 Data Preprocessing

In this step, the ASAI framework applies basic noise-removal methods to prepare the data for the next step. It extracts the image’s features by using CNN. Since images in a dataset can be of any size, while CNN needs a fixed-size input, the ASAI resizes any image to a fixed size, which is $224 \times 224 \times 3$ pixels, before feeding the image pixels to the CNN. We use a standard method to resize images, which is the bilinear interpolation method. For the surrounding text, we read HTML files from a dataset, extract the main article of the web pages, and remove all HTML tags. An article can have so many sentences, and processing too long word sequences with the

RNN may lead to training failure. Therefore, we limit the number of sentences that will be the input. In the experiment, we varied the number of sentences in an article from one to three. More sentences make the NN too heavy to train. The ASAI translates the surrounding text into a sequence of vocabularies by tokenizing every sentence. It counts the usage frequency of each word, any word that exists less than the threshold value is replaced by a UNK “unknown” word token, and the rest of the words are included in the word vocabulary. During the experiments, we vary the threshold value of infrequent words to find the optimal parameters of our model. By having this word vocabulary, every word in the surrounding text is encoded as an integer that indicates the word index in the created vocabulary.

3.2 Using VGG CNN as Feature Extractor

In this step, the CNN takes the fixed-size images from the previous preprocessing step. The ASAI reproduces deep CNN for object recognition, implemented and trained by Oxford’s Visual Geometry Group (VGG), which achieved remarkable performance on the ImageNet dataset [42].

The convolutional layers in VGG use a very small receptive field. 1×1 convolution filters act as a linear transformation of the input, which is followed by a Rectified Linear Unit (ReLU) as an activation function. The convolution stride is fixed to 1 pixel so that the spatial resolution is preserved after convolution. VGG has three fully connected layers: the first two have 4096 channels each and the third has 1000 channels, 1 for each class. All of VGG’s hidden layers use ReLU. VGG does not generally use Local Response Normalization (LRN), as LRN increases memory consumption and training time with no particular increase in accuracy. In the proposed model, VGG consists of 19 convolutional layers, activation layers with ReLU function, 5 max-pooling layers, and three fully connected (FC) layers at the end of the structure. The first two FC layers have an output dimension of 4096, and the third performs 1000-class classification for the ImageNet dataset. We removed the last layer output, since it is specific for the ImageNet dataset, and used the 4096-dimension output of the second-to-last FC layers as the image features in the ASAI framework. An NN learns its weights by a training algorithm, but some parameters specifying the NN structure need to be initialized. These include the number of layers used, the size of the filter, and the max-pooling layers to be allocated. We used the NN structure proposed by Simonyan et al. [9] and pre-trained with the ImageNet database [19]. It consists of 16 layers, which are combinations of convolutional, pooling, and FC layers. The last layer is a soft-max layer that maps the preceding output to ImageNet labels. These labels are the results of extracting information from image pixels. However, the second-to-last layer is used as a general feature extractor. Using this output, we can extract



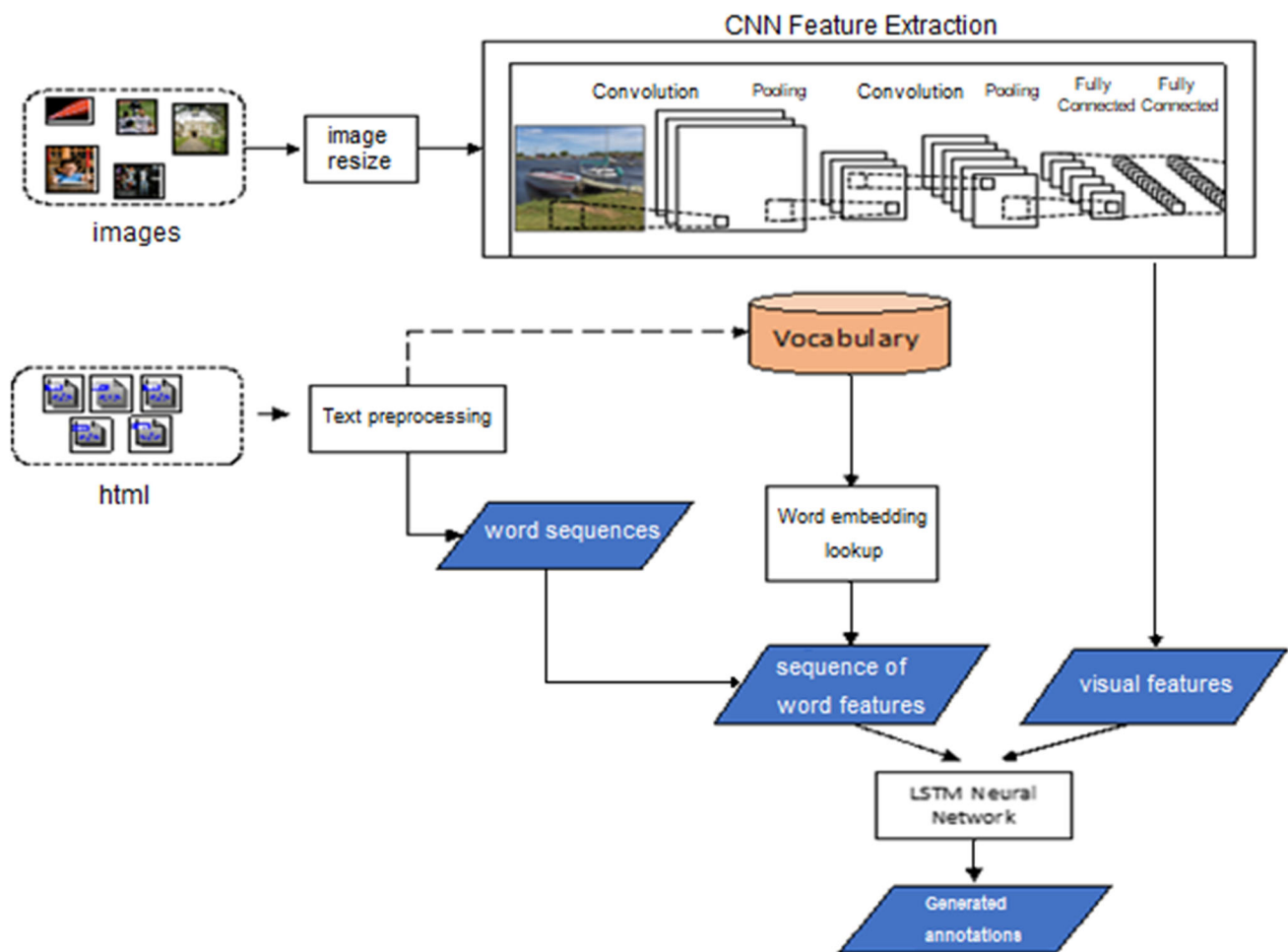


Fig. 1 Automatic semantic annotation of image framework

content-based features from other datasets and combine the features with different classifiers.

3.3 Sequence-to-Sequence Framework with Attention Mechanism

In this step, the ASAI framework uses RNN to orchestrate the word-embedding text features and the CNN image features to generate an image annotation. The RNN uses bidirectional LSTM (BiLSTM) cells [43]. The text and image inputs are regarded as a sequence of word features and image features, namely I , A_0 , A_1 , $\dots$, A_N . The model for this work follows a sequence-to-sequence framework [40]. The first part of the network is to encode the input sequence and obtain a fixed-sized state vector that represents the whole input sequence. The latter part is used to decode the output sequence from that vector as shown in Fig. 2.

In a sequence-to-sequence framework, the encoder part compresses all the inputs from x_0 to x_N into state vectors h_t and c_t . These state vectors will be the representation of the

input encoder and become the input states of the decoder. The state vectors may lose the main information from the original input. Therefore, an attention mechanism [44] is used to solve the bottleneck problem of the encoder-decoder model. The decoder part takes attention to each state in the encoder before generating a final output within a given time t . With an attention mechanism, we must look at all LSTM encoder states, from the state with initial input x_0 until x_N . Figure 3 illustrates the attention mechanism in a sequence-to-sequence framework.

To compute the hidden state y_t of attention decoder at each output time t , we define:

$$y_t = \text{concat}(d_t, d'_t) \quad (4)$$

$$d'_t = \sum_{i=1}^T a_i h_i \quad (5)$$

$$a_i^t = \frac{\exp(u_i^t)}{\sum_{i=1}^T \exp(u_i^t)} \quad (6)$$



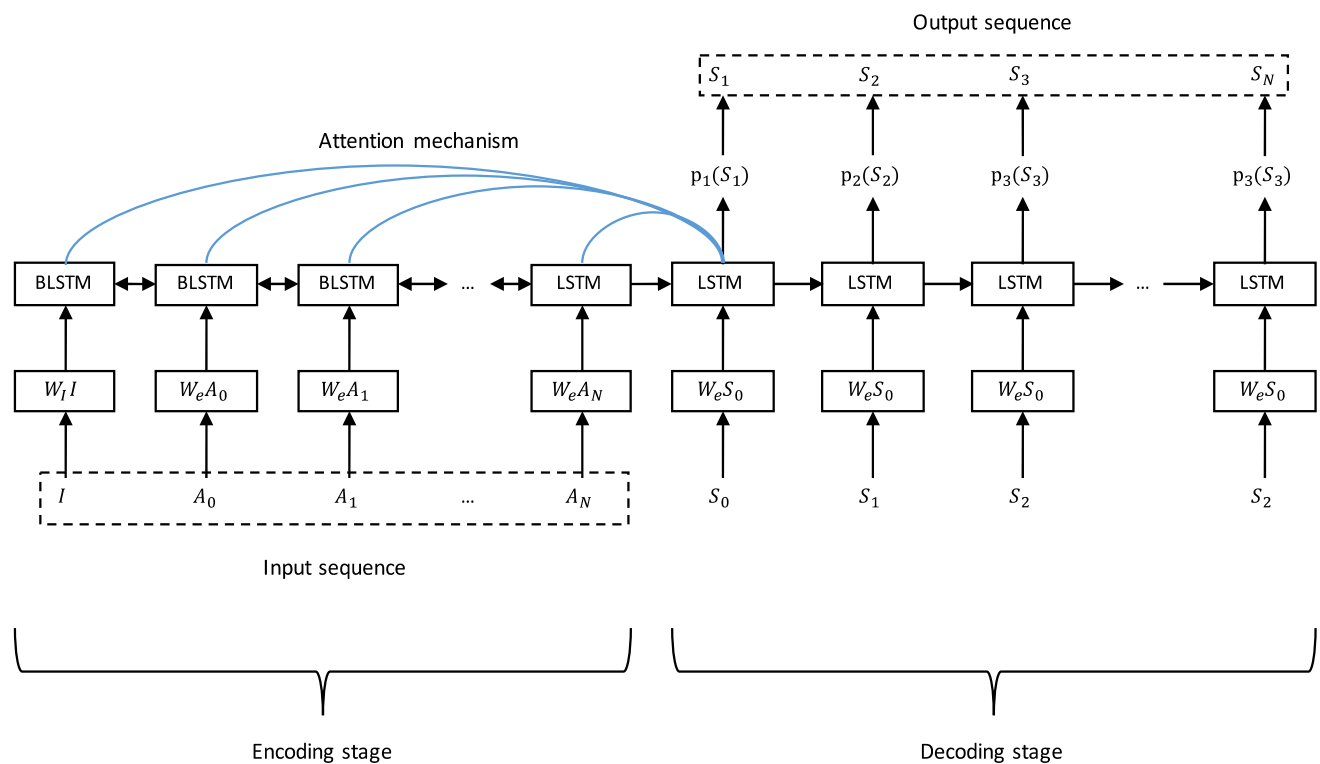


Fig. 2 LSTM-based RNN in ASAI framework

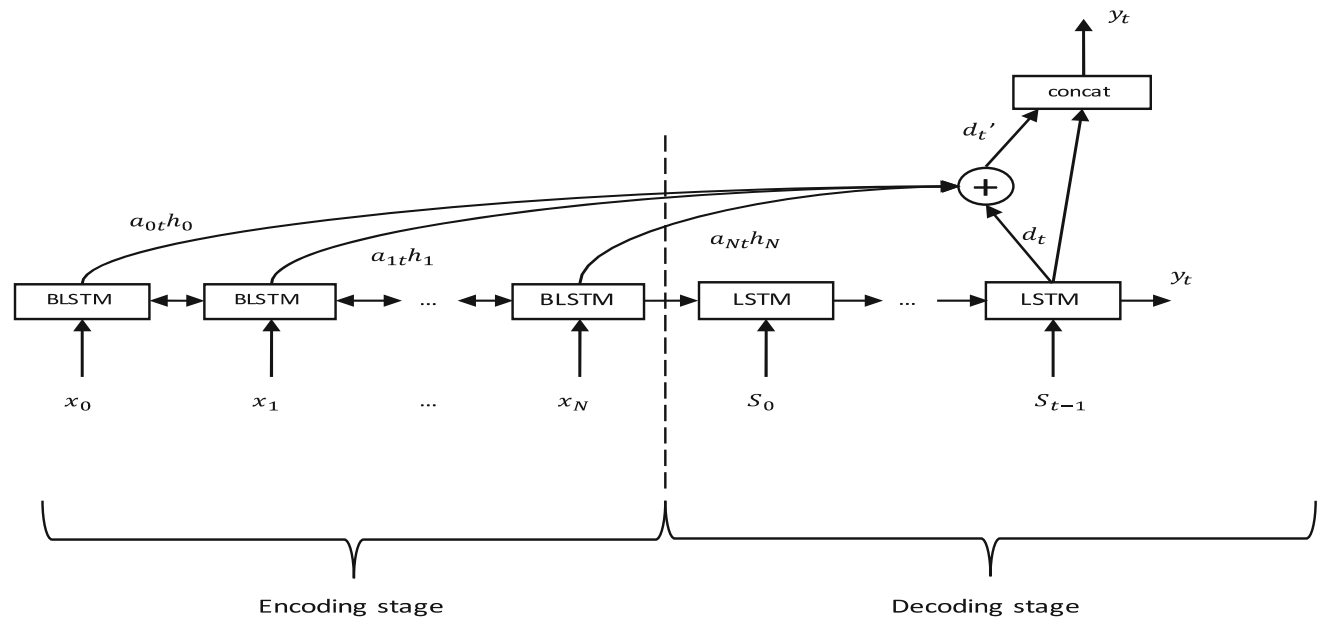


Fig. 3 Attention mechanism

$$u_i^t = v^T \tan h(W_1 h_i + W_2 d_t) \quad (7)$$

Overall, the LSTM-based RNN in the ASAI framework does the following. We take the first three sentences in the surrounding text and map every word in the sentence to a multi-dimensional vector. These three sentences were the

most weighted in the surrounding text. This was measured by counting the frequency usage of each word, and sentences with a word's count more than the threshold value were selected. Then we carry out word-embedding if the word occurs more than five times in the training data. Otherwise, we map the word to a vector word for a UNK word token.



After this process, we have a sequence of word features as a representation of the article, which is $A = A_0, A_1, \dots, A_{N-1}$. We insert the special “end of sequence” token at the end of article A such that $A = A_0, A_1, \dots, A_N$ where A_N is the word vector for the “end of sequence” token. Article A and the image features I from the CNN are arranged into a sequence input, such that $\text{input} = A_0, A_1, \dots, A_N, I$. We feed the LSTM encoder by the sequence input. All outputs of the LSTM encoder are ignored except the last state of the LSTM, which is the LSTM state when image feature I becomes the input. The last state vector is fed into the LSTM decoder at the first time step t_0 . For each time step t , the LSTM decoder is fed by the state of the previous time step. When in training, the decoder is fed by the ground truth or correct word feature of the previous time $t - 1$, and when in testing, the decoder is fed by the generated word of the previous time $t - 1$. For each time step t , the LSTM attention decoder output sequence will be multiplied by weights and fed into a soft-max function to generate the probability of word occurrence p_t in the given time step t . The word that has the highest probability becomes the output sequence of its corresponding time step. During training, we update the weights in the proposed model by minimizing a cost function concerning the parameters in the model, excluding the parameters in the VGG CNN. The cost function is given by:

$$J(S, A, I) = -\frac{1}{N} \sum_{t=1}^N \log p_t(S_t) \quad (8)$$

3.4 A New Dataset Preparation

Since the ASAI framework deals with both image and text modalities to describe images, in or to evaluate the effectiveness of the ASAI, we seek a dataset that meets the following criteria. It does not contain manual work. It does not rely on human involvement, so it can be scalable to a bigger dataset. It shows correspondences between text and data. The surrounding text can add both denotative and connotative information shown in the image. It has a huge amount of data samples, as the training requires much data training.

Manually building a dataset through human intervention to provide clean, high-quality, and reliable labels requires an expensive and laborious endeavor. This can also limit the number of data samples in the dataset. To overcome this issue, recent works on image annotation use crowdsourcing to build an extensive collection of data. Still, even with crowdsourcing, dataset building becomes too expensive. On the other hand, billions of images and their surrounding text appear on web pages and can easily be gathered using a web crawler. Therefore, we think that the framework should not rely on a dataset that involves much human involvement and should

be able to take advantage of the huge amount of data available on the Internet. Using easily obtainable data can be more scalable for future work when needing larger data. However, an automatically gathered dataset can be noisier than manually generated datasets. This framework should be able to handle this drawback.

We found existing datasets to support computer vision tasks. The available datasets for image annotation include ImageNet [19], PASCAL VOC 2012 [18], and Corel. These datasets are not suitable for our task since they do not provide text modality and are more suitable for image labeling without surrounding text features. Correspondingly, there are also datasets to support natural-language-description generations of images such as Pascal sentence [45], MS-COCO [46], Flickr8K [47], and Flickr30K [48] which contain 100 K, 8 K, and 30 K images, respectively, along with five sentences for each image that describe its visual content. Using these datasets, some recent works can generate image descriptions given an image [11, 23]. The task in the ASAI framework differs from the mentioned works. While these works use only the images modality as input to produce text description, the ASAI framework uses both text and image modality as input.

To the best of our knowledge, some datasets that are intended for generating semantic annotation from text and image are the BBC news dataset [31] and ION dataset [13]. In these datasets, image captions and surrounding text do not necessarily describe what the corresponding image depicts (denotative concepts). They can describe connotative concepts, i.e., there are no specific regions in the image depicting mentioned concepts in the text. BBC contains 3361 articles with images, but this number of data samples could be too small for training deep learning architecture, while the ION dataset contains 300 K articles and could be a better candidate. Unfortunately, the ION dataset cannot be used as it is, because the published features are different from what we are using, and we would need to compute the required features, while the original articles and images are not available due to copyright restrictions. Although a list of original URLs of websites is available, some URLs are already inaccessible, and we found that some websites do not contain complete article and image pairing. As a result, we consider that providing a new dataset is important for our goal and can be provided to other researchers. To build a dataset, we adopted and modified the existing ION dataset. This dataset contains 300 K articles containing images and the surrounding text from various news sources, i.e., New York Times, Daily Mail, Independent, etc. We decided to use the articles that are from Daily Mail only because other news sources need paid subscriptions for getting news data, and truncate articles that do not have the corresponding image. Some URLs are also inaccessible, so it turns out we only have 70 K pairs of articles and images. We downloaded the published source code in GitHub to gather the ION dataset and modified it. The source code is

Five radicalised footballers from east London who left the UK to join ISIS may hold the key to helping western authorities locate British executioner Jihadi John. The group of militants all moved from their native Portugal to London, where they converted to Islam before adopting extremist views and travelling to Syria to join the terrorist network's ranks. The men, who lived in Leyton and Walthamstow, have long been on the radar of intelligence officials - who believe the group plays a vital role in the production and dissemination of the sick beheading videos featuring Jihadi John. Scroll down for video In July 2014 - 39 days before James Foley became the first Western hostage to be murdered - one of the east London cell's ringleaders posted a message on Twitter indicating he had advance knowledge of the American journalist's grisly fate. Nero Saraiva, who lived in a one-bedroom council flat in Walthamstow for three years, tweeted: 'Message to America, the Islamic State is making a new movie. Thank u for the actors.' Just over a month later James Foley was beheaded in a YouTube video entitled A Message To America. European security officials now believe Saraiva and his East London cell are responsible for the filming and dissemination of the series of sickening beheading videos, including the execution of British hostages Alan Henning and David Haines. Experts hope the group's link to Jihadi John, who speaks with a London accent, will help them build a background profile of the ISIS executioner and his networks - and aid in any mission to capture or kill him, the Sunday Times reports. MI5 are said to have worked out the identity of Jihadi



when he crashed a high-powered classic sports car into a tree - just 100ft from the garage he stole it from. The crook and an accomplice broke into the property and were attempting a getaway when the black 1964 Mercedes-Benz 230 SL - worth £70,000 - smashed into a tree seconds after fleeing the property in Osielsko, north Poland. The dead burglar's accomplice drove off with a 1959



Mercedes-Benz 190SL - worth about £50,000. A man who was walking his dog nearby at the time said: 'I saw the car crash and ran over to see if I could help. The other car had driven off and when I looked inside the crashed car I saw a man slumped over the steering wheel wearing a balaclava and gloves. 'He was dead. I guess God was watching and decided to punish him.' The owner of the two Mercedes-Benz classic cars, Maciej Borkowski, thought the thief may have crashed because he was used to cars with power steering. He said: 'These are powerful old cars and they don't drive and steer like modern ones. You need to know what you're doing.' A police spokesman said: 'We are trying to identify the driver which is proving difficult as we have no record of him in our files and no one has come forward to claim the body. 'He is a bit of a mystery.' But one local added: 'The bigger mystery is how his accomplice is going to sell the car now, as everyone will know it was stolen and that someone died in

Fig. 4 Two examples of web pages in the created dataset

written in Java, and we modified it to do the following. For each web page in the original ION dataset, the code checks whether the web page includes an image or not. If there is no image on the web page, then it is ignored. If the website has an article with more than three sentences, then it includes it in the dataset. We do not use websites that have articles of less than three sentences, because we assume very short articles do not add additional information to the images. The filtered dataset contains 76,324 data samples from the Daily Mail website. The filtered dataset is divided into 90% for training, 5% for validation, and 5% for testing. In other words, it has 68,692 data samples for training, 3816 data samples for validation, and 3816 data samples for testing. Figure 4 shows two examples of data samples in the generated dataset.

4 Experimental Results and Discussion

In this section, we assess the effectiveness of the ASAI framework which we implemented using Python code with TensorFlow library [49] on NVIDIA Tesla K20 6 GB GPU. In all experiments, we use our dataset and use Cross-Entropy loss as a cost function (known as loss function) and the Adaptive Moment Estimation (ADAM) training algorithm [50] to minimize the loss function. As mentioned in Eq. (8), the loss function is the negative log of the probability of generating the correct sentence given the image and text features, and we used the loss function as one of the performance metrics with the logarithmic operation to prevent computational round-off errors. If the model has an output that is identical to the training sentence caption, we will not get zero loss since we have regularization. Otherwise, the loss will be a very high number if the output model has no identical sequence. Therefore, a lesser loss function value means the model has

better performance. The training process stopped whenever the validation loss was not decreasing with training iteration, regardless of the training loss decrement. The learning rate can be set relatively high to speed up the training and reduce the risk of getting a local minimum. However, it should be reduced at each iteration to prevent bouncing, i.e., the loss will keep oscillating and never get the minimum value. Therefore, during training, the learning rate is firstly set to 5×10^{-4} and then reduced by 5% at each epoch. Figure 5 shows an example of the training process of the model, and the best model is at epoch 28 because it has the lowest validation loss. The x-axis in the graph shows the number of iterations, while the y-axis shows the respective cost value.

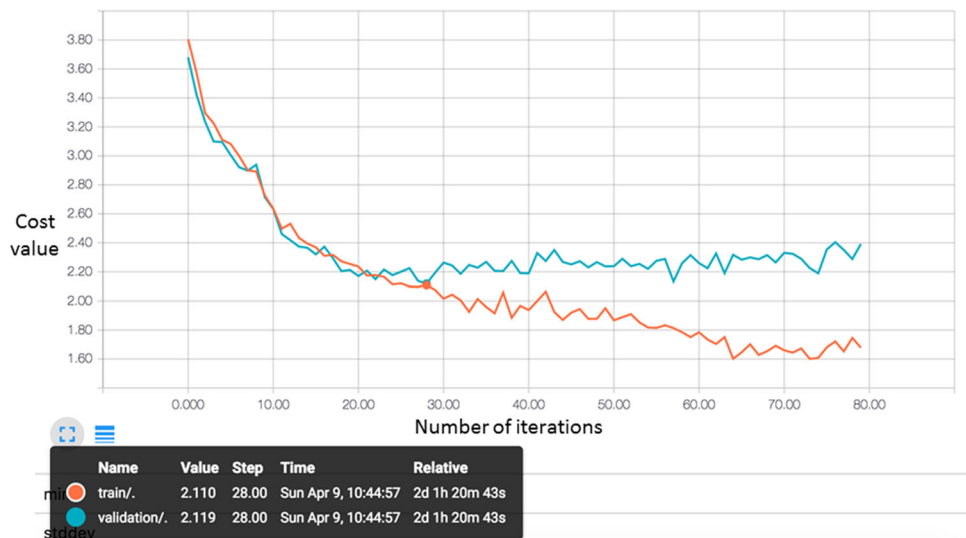
As for the evaluation metrics to evaluate the system performance, we used BLEU-N [51] and the loss function scores stated in Eq. (8). The higher the BLEU-N score, the better the performance will be, while the lower loss function value indicates the better performance.

4.1 Hyper-Parameters Set-Up

Several sentences per article and word count threshold: The number of sentences per article indicates how many sentences are kept from that article and the word count threshold controls the addition of a word into the word-embedding vocabulary dictionary. Any word in the training data with a count less than the threshold value will be replaced by the special word "UNK." Since a high threshold will make important words not present in the dictionary, while a low threshold will mean that too many words will be in the dictionary leading to bigger memory requirements and slower training, we varied the vocabulary limit and sentence count per article during the hyper-parameters tuning. We cannot use more than three



Fig. 5 An example of the training process of the ASAI framework



sentences for each article, because it requires too much memory, which will make the system unable to train. Intuitively, lowering the word count threshold and adding the number of sentences per article will make the vocabulary size bigger. Therefore, too many words may make the system overfitting and hard to train, leading to higher validation loss and decreasing BLEU performance. On the other hand, setting the threshold too high and using fewer sentences per article would make the system unable to generate sentences with rich vocabulary resulting in a detrimental effect on the system performance. We set the number of sentences per article to three, and the word count threshold to seven, because this is when the system achieves its highest performance with a vocabulary size of 23,580, which is somewhere between the highest number of vocabulary size (42,129) and the lowest (14,643).

Several RNN Layers: In general, we can compose an RNN cell with multiple LSTMs to increase the model performance. However, this method may not be effective in several applications, so we need to verify empirically if increasing the number of RNN layers is beneficial or not. In our experiments to find good hyper-parameters, all systems with one RNN layer outperformed all the other systems with two RNN layers. Thereby, we set the number of RNN layers to one.

LSTM size: LSTM size is the number of output vector dimensions of the LSTM cell. Using more units can reduce training set loss. We found that the system with an LSTM hidden size of 128 has the least validation loss value among other systems with lower LSTM hidden sizes. Therefore, the LSTM networks with a higher LSTM size tend to perform better, and that is why we set the number of neurons of LSTM to 128.

Batch size: Batch size controls how much-processed data is sampled at each iteration. A large batch size tends to

Table 1 BLEU-1 evaluation of image caption predictions on Flickr 8K and Flickr 30K

Model	BLEU- 1 ON Flickr8K	BLEU- 1 ON Flickr30K
Mao et al. [52]	0.58	0.55
Google NIC et al. [24]	0.63	0.66
Karpathy et al. [33]	0.58	0.57
ASAI without text input	0.61	0.59

improve GPU parallelization, but the memory size of GPU limits the batch size. We set the batch size to 24 for all experiments to ensure that the GPU memory suffices during training. The running program would stop because of insufficient GPU memory when we increased the batch size to more than 24.

4.2 Performance Comparison with Existing Work

Most existing works use datasets that do not include surrounding text for each sample, such as Flickr 8K and Flickr 30K. So, when using these datasets, the ASAI framework is identical to the Neural Image Caption (NIC) framework proposed by Vinyals et al. [24], as NIC uses only image features as input to generate text description. We reproduced the NIC framework proposed by Vinyals et al. [24] as a baseline. Image input is resized into a fixed size of 224×224 and fed into the CNN. The image features from CNN are then used by LSTM neural networks to get a sentence description of the image. The Table 1 shows the comparative performance of our reproduced model of NIC and others on Flickr 8K and Flickr 30K datasets. We do not achieve the exact performance mentioned in their paper, because we might have had

Table 2 A comparison of the ASAI framework with other models

Model	Val- loss	BLEU- 1	BLEU- 2	BLEU- 3
NIC	3.105	27.91	3.84	0.60
Text summarization	2.273	15.54	5.48	3.08
ASAI	2.243	27.57	9.98	5.87
BreakingNews	–	19.6	8.9	1.6

some minor implementation differences from the original work, such as how they preprocessed the data before putting the data into neural networks. However, it still outperforms other work results such as Karpathy et al. [33]. Table 2 shows a comparison of the ASAI framework's performance evaluation with other models, i.e., NIC, Rush et al. [38, 39].

From Table 2, the ASAI framework has the least validation loss (2.243) and a better BLEU score (BLEU-2 of 9.98 and BLEU-3 of 5.87), compared to the system that uses image features only (NIC) and the system that uses text features only (Text summarization). We see that using both image and text modality can help the effectiveness of the framework, indicated by the lower validation loss value. The validation loss of the framework can be minimized even more by using both image and text features, compared to using text or image features only.

In addition, we also compute BLEU 1-g and 2-g scores [51] to evaluate the results, shown in Table 2. BLEU is one of the standard automatic evaluation methods of machine translation; it is the modified version of the precision measurement in information retrieval. The model that uses only image features (NIC framework) can achieve a high BLEU-1 score (27.91), but low BLEU-2 and BLEU-3 (3.84 and 0.60). One reason is that NIC generates short sentences, and the BLEU score has a drawback of preferring shorter sentences. In addition, NIC also gives many meaningless short sentences, and the BLEU is increased from very common words like “the”, “was”, and so on. The ASAI framework can achieve better meaningful sentences, indicated by more BLEU-2 and BLEU-3 scores. We also include the results from Ramisa et al. [1]. They proposed a similar image caption generator and the Breaking-News dataset. Although we do not use the same dataset, the Breaking-News dataset has similar characteristics to our dataset, i.e., both datasets are extracted from news websites where the surrounding text may have irrelevant information to the image. Also, the captions may not necessarily describe salient objects that appeared in the image. Table 2 shows that the ASAI framework has a higher BLEU score than Breaking-News.

4.3 Manual Evaluation

Next, we evaluated the results using manual evaluation. However, since manual evaluation is expensive, we only sample

Table 3 Manual evaluation results

Score	ASAI	NIC
1	7	27
2	10	3
3	7	0
4	6	0
Average	2.42	1.1

31 data (decided randomly) in the testing dataset. We assess the output by classifying the captions into four parts:

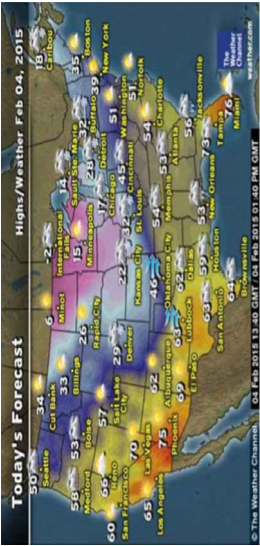
- Not describing the image at all
- Somewhat related to image
- Describing the image with much noise
- Describing the image with low noise

The Table 3 shows the manual evaluation results, and Table 4 shows some test samples of the ASAI framework output. Table 4 shows that ASAI can generate image annotations, even with the least relationships with the images and their surrounding text. For example, the last sample in the table shows a person on a mountain in New Zealand. The annotation can mention the concept “New Zealand” but fails to mention other concepts correctly. Therefore, since the results still contain errors, they may not be comparable to manual human annotations. The automatic image annotation task in news images is very challenging even for humans because it requires awareness and knowledge of the specific news events.

The errors in generated sentence annotations may be caused due to several reasons. First, there are always words that do not have many occurrences in the training dataset. The model does not have enough training to learn using uncommon words. Consider the example in Table 4 where the ASAI framework cannot mention the word “Dino 246GT”, which has no occurrence in the training dataset and is replaced by a UNK word token. Second, the framework only considers the first of several sentences of the articles for extracting surrounding text information. While the attention mechanism helps the RNN to extract the important information from the input sequence, it cannot process the whole article, because



Table 4 Sample of ASAI framework annotation results

Image and output	Surrounding text
	For the second time this week, a winter storm was barreling towards the Northeast, but thankfully looked set to dump a relatively small amount of snow. From late on wednesday through to thursday night, snowfall will ramp up, bringing up to six inches of snow in eastern Vermont, New Hampshire, and Maine
	A woman armed with a knife stole an ambulance and later crashed it into a big rig, killing herself and a truck driver, according to police. The woman, who has not been named, carjacked the ambulance and sped away after paramedics responded to a request for medical aid at a home on North End Drive, Frazer Park, California

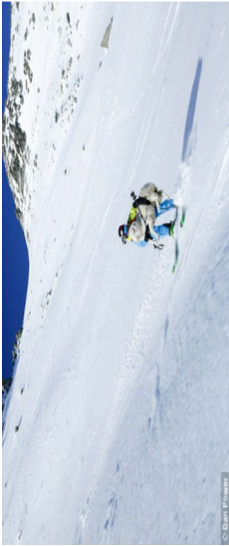
Output: frightening: the winter storm was #UNK# off the northeast, New Hampshire, in eastern Vermont

Score: 4

Output: crash: a woman, who has not been named, left the ambulance and sped away after paramedics responded to a request for medical aid at a home on north end drive, #UNK# park, where he was injured

Score: 3

Table 4 (continued)

Image and output	Surrounding text
	A yellow Ferrari bought by Elton John when he first became famous in 1972 has gone on sale for £300,000. The Ferrari Dino 246GT, one of fewer than 500, belonged to the singer for six months before he gave it to the longstanding drummer, Nigel Olsson
<i>Output:</i> a yellow Ferrari #UNK#, one of fewer than 500 belonged to the singer for six months before he gave it to silly #UNK# Score: 3	
	“It’s a good yarn”, that’s certainly one to be re-told by pro skier turned sheep rescuer, Pete Oswald. The 29-year-old was on Hector Mountain, south of New Zealand’s South Island when he noticed a little bundle of wool snowballing down the slopes which then landed 100 m away from him
<i>Output:</i> the golden mountain, south of New Zealand’s South Island, was on exercise mountain, south of New Zealand’s South Island when he noticed it of photographing beggars , #UNK# down the lawn that then landed 100 m away from Score: 3	

it increases the trainable parameters rapidly as the number of the processed sequence in the RNN is increased. Thus, it requires too much memory. Therefore, if the main information is in the middle of the article, the system will not get the information.

4.4 Converting Free-Text Annotations to RDF Annotations

The generated image annotations from the ASAI framework are in free-text form. This form can be further processed into more structured metadata such as Resource Description Framework (RDF). RDF is a standard way of writing metadata in a directed graph data model to describe any web resources, including images. RDF metadata is expressed by a triple of the forms: subject, predicate, and object. We can use metadata from an ontology (which is a set of graphs that contains relationships between concepts) and create links between concepts to describe an image. RDF annotations can help image retrieval because they can improve precision by disambiguating terms like homonyms and can extend the query to bigger recall by associating synonym and hypernym terms. We demonstrate how the generated semantic annotation can be converted to RDF metadata format using some NLP parsers and basic information extraction methods.

To convert a plain text to structured RDF triples, we used Stanford parser and Senna parser. Stanford dependency parser breaks down a text into a parse tree that shows the syntactic structure and grammatical relationships among words. Each word is tagged with its part-of-speech tags such as noun, verb, subject, modifier, object, etc. SENNA framework performs a semantic role labeling task. The task is to detect and classify word arguments that are associated with the verb of a sentence. For example, given the sentence “Ahmad is eating an apple”, the task is to recognize “to eat” as the predicate, “Ahmad” as the first argument classified as “agent” or “eater”, and “apple” as the second argument classified as “entity being eaten”. Unlike syntactic information, semantic information is at a higher level as it may not depend on the order of words in the sentence.

Having the output from the Stanford parser and SENNA framework, we can create RDF triples annotations by following these steps. SENNA gives numbered arguments to words that have relationships to a verb. From this output, we create RDF triples with the format: (word, has-role, arg) for each word. The numbered arguments are converted to corresponding meanings, namely “agent” for arg0 and “patient” for arg1. From Stanford parser output, we obtained a list of words having direct relations that expose important structures of sentences such as objects and subjects. For each detected verb, a subject list and an object list are created. Every word that has subject-role dependency such as a nominal subject (nsubj), or clausal subject (csubj) will be

put into the subject-list, whereas any other word that has any other dependency will be put into the object-list. RDF triples are created by making a mesh connection between words in the subject list, the verb, and words in the object list. Any concept in those lists is associated with a corresponding concept that is defined in DBpedia ontology [53], so anyone can request further information about any concept appearing in the image, including its synonyms and hypernyms. Figure 6 shows an example of an RDF annotation from the free-text description of an image generated by the ASAI framework.

5 Conclusion and Future Work

The contribution of this paper is to generate semantic image annotations by taking advantage of the information contained in the image’s pixels and the surrounding text using new techniques, namely convolutional neural networks (CNN) and RNN. We have proposed and developed the ASAI framework to generate semantic annotation of images in the form of sentences and RDF. The proposed framework was evaluated using a dataset based on images and text created from the ION dataset. The process of combining an image’s low-level features and surrounding text is carried out using RNN with LSTM cells. The used feature extraction methods are data-driven and do not rely on the human expert domain; therefore, this framework could be applicable in wider fields. The experiments show promising results and demonstrate the effectiveness of the proposed framework. It is also worth mentioning that the ASAI framework works with a dataset that can be easily gathered from the Internet without utilizing much time and human resources. As machine intelligence is not comparable with human work in general, we are aware that the result is nowhere close to perfect compared to manual human annotation. Therefore, for future work, the framework could be explored with different datasets. The datasets can be in a non-English language, and the topics in the datasets could be more specific such as sport-related, culture-related news, etc. The used dataset could be larger to improve the performance. Moreover, advanced preprocessing methods can be further explored, since we only implemented the basic preprocessing methods. Other feature-extraction techniques could be added, i.e., part-of-speech (POS) tags, named entity tags, and the TF-IDF of each word in addition to its word-embedding features.

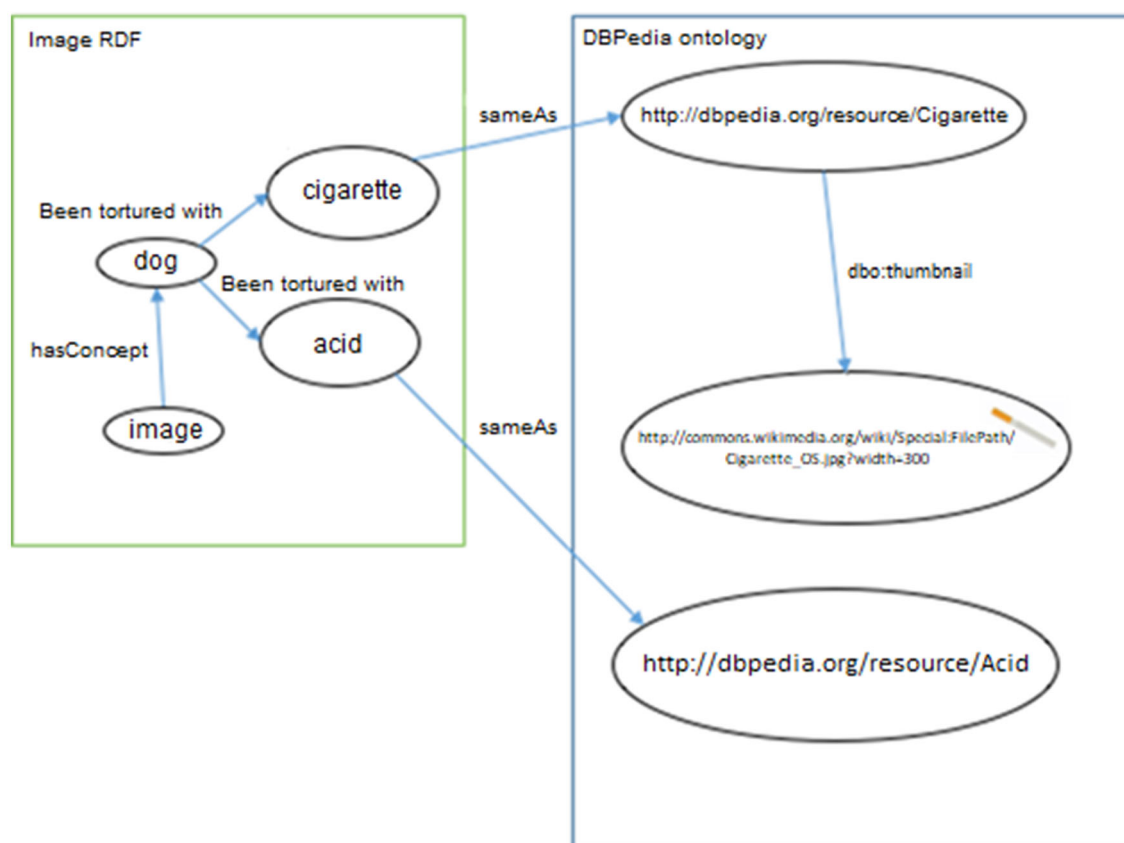


Fig. 6 RDF annotations of a sample image generated by the ASAI framework



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